

Grover's quantum searching algorithm is optimal

Christof Zalka*
zalka@t6-serv.lanl.gov

November 26, 2024

Abstract

I show that for any number of oracle lookups up to about $\pi/4 \sqrt{N}$, Grover's quantum searching algorithm gives the maximal possible probability of finding the desired element. I explain why this is also true for quantum algorithms which use measurements during the computation. I also show that unfortunately quantum searching cannot be parallelized better than by assigning different parts of the search space to independent quantum computers.

1 Quantum searching

Imagine we have N cases of which only one fulfills our conditions. E.g. we have a function which gives 1 only for one out of N possible input values and gives 0 otherwise. Often an analysis of the algorithm for calculating the function will allow us to find quickly the input value for which the output is 1. Here we consider the case where we do not know better than to repeatedly calculate the function without looking at the algorithm, e.g. because the function is calculated in a black box subroutine into which we are not allowed to look. In computer science this is called an oracle. Here I consider only oracles which give 1 for exactly one input. Quantum searching for the case with several inputs which give 1 and even with an unknown number of such inputs is treated in [4].

Obviously on a classical computer we have to query the oracle on average $N/2$ times before we find the answer. Grover [1] has given a quantum algorithm which can solve the problem in about $\pi/4 \sqrt{N}$ steps. Bennett et al. [3] have shown that asymptotically no quantum algorithm can solve the problem in less than a number of steps proportional to $\sqrt{N}$. Boyer et al. [4] have improved this result to show that e.g. for a 50% success probability no quantum algorithm can do better than only a few percent faster than Grover's algorithm. I improve

*Supported by Schweizerischer Nationalfonds and LANL

the proof, showing that for any number of oracle lookups Grover's algorithm is exactly (and not only asymptotically) optimal.

The abovementioned proofs have shown that asymptotically proportional to $\sqrt{N}$ steps are necessary for quantum searching. They have not said whether these steps can only be carried out consecutively or whether they could (partially) be done in parallel. If they could be done in parallel then a quantum computer containing S oracles (thus S physical black boxes) running for T time steps could search a search space of $O(S^2T^2)$. Now any unstructured search problem can simply be parallelized by assigning different parts of the search space to independent search engines (whether quantum or classical). But using this "trivial" parallelization we can only search a search space of $O(ST^2)$ using S independent quantum computers running Grover's algorithm. I show that we can not do better.

Grover's quantum searching algorithm distinguishes between the N possible one-yes oracles (yes = 1) using $\pi/4 \sqrt{N}$ oracle calls (that is, evaluations of the function). It makes the same sequence of operations $\pi/4 \sqrt{N}$ times. The sequence consists of 4 simple operations. The input state to the algorithm is the (easily constructed) uniform amplitude state:

$$\text{initial state} \quad \phi_0 = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} |x\rangle \quad (1)$$

The $|x\rangle$ are the $N = 2^l$ computational basis states (where every one of the l qubits is either 0 or 1) which correspond to the possible inputs to the oracle. Thus for Grover's algorithm N has to be a power of 2. Again [4] have generalized this to arbitrary N .

The 4 operations of Grover's algorithm are then:

1. $|y\rangle \rightarrow -|y\rangle$ for the one marked y
2. H^l
3. $|x\rangle \rightarrow -|x\rangle$ for all $x \neq 0$
4. H^l

The first step is really the invocation of the oracle. The input we are looking for is y . An oracle giving 0 or 1 can easily be changed into an oracle which conditionally changes the sign of the input. This can be done by preparing the qubit into which the oracle output bit will be XORed in the state $(|0\rangle - |1\rangle)/\sqrt{2}$. Of course the oracle will need work space, but as we expect these work qubits to be reset to their pre-call value after each oracle call, we do not really have to care about them in the quantum algorithm as they "factor out" (after the oracle call they form a tensor product with the rest of the quantum computer). The requirement that the work qubits have to be "uncomputed" means that

the algorithm in the oracle may take longer than its conventional irreversible version.

The second step applies a Hadamard transform to every one of the l qubits (shorthand H^l). The Hadamard transform is given by the following matrix:

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad (2)$$

The third step changes the sign of all computational basis states except for $|0\rangle$ and the forth step is the same as the second.

The following results are straightforward to obtain: After any number i of iterations of these 4 operations the state of the QC can be written as a linear combination of two fixed states:

$$\phi_i = A_i \frac{1}{\sqrt{N-1}} \sum_{x \neq y} |x\rangle + B_i |y\rangle \quad (3)$$

One application of the 4 operations gives:

$$\begin{aligned} A_{i+1} &= \left(1 - \frac{2}{N}\right) A_i - 2 \frac{\sqrt{N-1}}{N} B_i \\ B_{i+1} &= 2 \frac{\sqrt{N-1}}{N} A_i + \left(1 - \frac{2}{N}\right) B_i \end{aligned}$$

This is simply a SO(2) rotation with

$$\cos(\varphi) = 1 - \frac{2}{N} \quad \text{and} \quad \sin(\varphi) = 2 \frac{\sqrt{N-1}}{N} \quad (4)$$

Thus $\varphi \approx \sin \varphi \approx 2/\sqrt{N}$ and therefore after $\pi/4 \sqrt{N}$ steps we obtain a state very close to $|y\rangle$.

It turns out (and is easy to check) that the initial (uniform amplitude) state can be written in terms of half the above angle:

$$\phi_0 = \cos(\varphi/2) \frac{1}{\sqrt{N-1}} \sum_{x \neq y} |x\rangle + \sin(\varphi/2) |y\rangle \quad (5)$$

Thus the success probability after T oracle calls is exactly

$$p_T = \sin^2(T\varphi + \varphi/2) \quad . \quad (6)$$

(Note that after about $\pi/4 \sqrt{N}$ iterations of Grover's algorithm the success probability goes down again!)

A noteworthy remark: Actually because towards the end the success probability goes very slowly to 1, if we want to minimize the average number of

steps¹, it pays off to end the computation earlier and run the risk to have to start over. A simple calculation shows that the average number can thereby be reduced by some 12.14% relative to the above result.

2 tight bound on quantum searching

Here I sketch my version of the proof from [4] which gave the tightest limit on quantum searching so far. It is an extension of the earlier $\Omega(\sqrt{N})$ proof in [3]. Later I obtain my results by further refining the same proof.

In the proof I assume a quantum computation consisting only of unitary transformations (plus the final measurement) without measurements during the computation. This can be done without loss of generality: Clearly a measurement of a qubit whose outcome will not be used to make decisions on what further unitary transformations should be applied, can be delayed to the end. For a practical QC it seems likely that what further unitary transformations will be applied will depend on outcomes of intermediary measurements, like e.g. in error correction. Thus we will probably have a “hybrid” quantum-classical computer, where the classical part reads measurement outcomes and depending on that, controls the exterior fields that induce unitary transforms on the qubits. The point now is that in principle the classical part can simply be replaced by quantum hardware which does the same. This may use more space as we are now restricted to reversible computation, but for what concerns us, it of course does not increase the number of oracle invocations, and this is really all we care about here.

So we do not care about the cost of any other unitary transforms, actually we do not even ask whether they can efficiently be composed of elementary gates. Of course once we have established that even under this general viewpoint Grover’s algorithm is optimal, we know that the “auxiliary” unitary transforms can be realized with just a few elementary gates. Actually I expect that in any sensible application of Grover’s algorithm these auxiliary operations are going to be much easier (and faster) than the oracle call.

The proof gives a limit on the success probability achievable with T (for time) oracle lookups. Thereby we average over the N possible oracles. In computer science one is usually interested in the worst case, that is the oracle for which the success probability is the smallest. Because the worst case probability is smaller or equal to the average case probability we also get an upper limit on the former. In Grover’s algorithm the success probability is independent of the oracle so the worst case and average case probabilities are the same and thus Grover’s algorithm is also optimal for the worst case probability.

The proof works by analyzing how the difference of the QC states between the cases when we have a specific one-yes oracle and when we have the empty oracle (always giving 0) evolves. For the empty oracle case I denote the QC

¹as opposed to the maximum (worst case) number of steps

state after i oracle invocations by ϕ_i whereas ϕ_i^y denotes this state when we have an oracle that gives yes only for input y . More precisely, these are the QC states just before the next oracle call, thus in one register of the QC there must be the input to the oracle.

The proof consists of two parts. The central part of the proof gives a bound on how far from the empty-oracle case the state can have diverged after T oracle calls when the oracle has one yes. To get a meaningful statement we have to average over all possible one-yes oracles, as for any given y a special algorithm could be made that would do especially well for this case. The statement is:

$$\sum_{y=0}^{N-1} |\phi_T^y - \phi_T|^2 \leq 4T^2 \quad (7)$$

The second part of the proof gives an upper bound on the success probability p in terms of the left hand side of the above equation:

$$2N - 2\sqrt{N}\sqrt{p} - 2\sqrt{N}\sqrt{N-1}\sqrt{1-p} \stackrel{1.}{\leq} \sum_{y=0}^{N-1} |\phi_T^y - \phi_T|^2 \quad (8)$$

Where the “1.” above the “ $\leq$ ” is for later discussion. I prove this inequality in the appendix, as it is not central to the understanding of the proof.

Both inequalities together then give the desired lower bound on T in terms of N and p . Asymptotically and for $p = 1$ the statement is $T \geq \sqrt{N/2}$. I now derive the “central” inequality 7. To simplify the notation I assume that, like in Grover’s algorithm, in every iteration (each containing one oracle invocation) the quantum computer makes the same sequence of operations. It is easy to see that the proof works just as well without this restriction (just add additional indices to U and U_y). By

$$\Delta U = U - U_y \quad (9)$$

I denote the difference between the unitary transformation corresponding to the empty oracle and the unitary transformation corresponding to the oracle giving 1 only for input y . The transformations U and U_y will act identically on all computational basis states except those where the register holding the input to the oracle is in state y . To get an upper bound on $|\phi_T^y - \phi_T|$, consider the following:

$$\phi_T = (U_y + \Delta U)\phi_{T-1} = U_y(U_y + \Delta U)\phi_{T-2} + \Delta U\phi_{T-1} = \dots \quad (10)$$

$$= \phi_T^y + \sum_{i=0}^{T-1} (U_y)^i \Delta U \phi_{T-1-i} \quad (11)$$

Then

$$\begin{aligned}
|\phi_T^y - \phi_T| &= \left| \sum_{i=0}^{T-1} (U_y)^{T-1-i} \Delta U \phi_i \right| \stackrel{2.}{\leq} \sum_i |(U_y)^{T-1-i} \Delta U \phi_i| = \\
&= \sum_i |\Delta U \phi_i| \stackrel{3.}{\leq} 2 \sum_i |P_y \phi_i|
\end{aligned}$$

Where P_y is the projector onto those computational basis states which are going to query the oracle on input y . The numbering of the inequality signs is again for later discussion. For the next step I need the inequality $(\sum a_i)^2 \leq T \sum a_i^2$, where the a_i 's are any T real numbers. It follows from the equality

$$\left(\sum a_i \right)^2 + \frac{1}{2} \sum_{i,j} (a_i - a_j)^2 = T \sum a_i^2 \quad (12)$$

which is easy to verify. So now we get:

$$|\phi_T^y - \phi_T|^2 \leq \left(2 \sum_i |P_y \phi_i| \right)^2 \stackrel{4.}{\leq} 4T \sum_i |P_y \phi_i|^2 \quad (13)$$

By summing this over all y 's we get:

$$\sum_{y=0}^{N-1} |\phi_T^y - \phi_T|^2 \leq 4T \sum_i \underbrace{\sum_y |P_y \phi_i|^2}_{=1} = 4T^2 \quad (14)$$

3 improving this bound

To see how tight the above inequality is, let us look at the 4 (numbered) inequalities in the proof which are then concatenated to yield the final inequality. Let us see how well Grover's algorithm (after any number of steps and for any marked y) does on these 4 inequalities. It turns out that it saturates all but the 2. inequality. For the 1. inequality this is shown in the appendix. The 3. (in)equality is easily verified. It is true because in the first of Grover's 4 operations the sign of $|y\rangle$ is changed, thus maximizing the distance between $|y\rangle$ and $-|y\rangle$. The 4. inequality is saturated because for Grover's algorithm the ϕ_i 's are all identical. So let us now concentrate on the 2. inequality:

$$|\phi_T^y - \phi_T| = \left| \sum_i (U_y)^{T-1-i} \Delta U \phi_i \right| \stackrel{2.}{\leq} \sum_i |(U_y)^{T-1-i} \Delta U \phi_i| \quad (15)$$

For Grover's algorithm, U is the identity and thus $\phi_i = \phi_0$, so

$$\Delta U \phi_i = 2P_y \phi_i = \frac{2}{\sqrt{N}} |y\rangle \quad (16)$$

As mentioned before, U_y just carries out a $\text{SO}(2)$ rotation on the space spanned by $|y\rangle$ and $1/\sqrt{N-1} \sum_{x \neq y} |x\rangle$. Thus for Grover's algorithm the vectors in inequality #2 do not all point in the same direction, rather if drawn one after the other (to form the vector sum) they form an arc. This prevents the inequality from being saturated and explains the discrepancy of the tight bound in [4] from the performance of Grover's algorithm.

So let us try to find a tighter (upper) bound on $|\phi_T^y - \phi_T|$. To this end I write equation 10 a little bit differently (where of course T does not mean "transpose"!):

$$\begin{aligned} \phi_T = (U \phi_{T-1} - U_y \phi_{T-1}) &+ (U_y U \phi_{T-2} - U_y U_y \phi_{T-2}) + \dots \\ &\dots + ((U_y)^{T-1} U \phi_0 - (U_y)^T \phi_0) + \phi_T^y \end{aligned}$$

This has the form

$$\psi_0 - \psi_T = (\psi_0 - \psi_1) + (\psi_1 - \psi_2) + (\psi_2 - \psi_3) + \dots + (\psi_{T-1} - \psi_T) \quad (17)$$

where all ψ_i 's are normalized. The question now is how we have to choose $\psi_1, \psi_2, \dots, \psi_{T-1}$ in order to minimize $\sum |\psi_i - \psi_{i+1}|^2$ when ψ_0 and ψ_T are fixed. Note that the relative phases of different states of the QC have no physical meaning and are therefore only a matter of convention. This is because quantum states are really given by 1-dimensional subspaces (rays) of a Hilbert space and not by vectors. We can thus assume that $\langle \psi_0 | \psi_T \rangle$ is real and non-negative.² Intuitively it is clear that for the minimum the ψ_i 's have to be evenly spaced along the arc between ψ_0 and ψ_T . (Note that this is obviously the case for Grover's algorithm!) Formally this can be established by setting the derivative with respect to the components of ψ_i equal to zero, which yields:

$$\psi_i = \frac{\psi_{i-1} + \psi_{i+1}}{|\psi_{i-1} + \psi_{i+1}|} \quad (18)$$

We can imagine all these vectors to lie in a 2-dimensional real vector space spanned by ψ_0 and ψ_T . From planar trigonometry we then get (draw a picture with a line bisecting the angle between ψ_0 and ψ_T):

$$|\psi_0 - \psi_T|^2 = \left(2 \sin\left(\frac{\alpha}{2}\right)\right)^2 \Rightarrow |\psi_i - \psi_{i+1}|^2 = \left(2 \sin\left(\frac{\alpha}{2T}\right)\right)^2 \quad (19)$$

²Actually it would be nicer to write the proof in terms of absolute values of scalar products only, thus avoiding unphysical quantities like the difference of state vectors

Where α is the angle between ψ_0 and ψ_T and α/T the angle between ψ_i and ψ_{i+1} for all i .

So we now have that (compare to equation 13)

$$|\phi_T - \phi_T^y|^2 \leq f \left(4 T \sum_i |P_y \phi_i|^2 \right) \quad \forall y \quad (20)$$

where $f(x)$ describes the improvement in our bound. It is given by (T is a fixed parameter)

$$f \left(\underbrace{4 T^2 \sin^2 \left(\frac{\alpha}{2T} \right)}_{\approx \alpha^2} \right) = \underbrace{4 \sin^2(\alpha/2)}_{\leq \alpha^2} \quad (21)$$

We now want to sum equation 20 over all y 's. We use calculus to get an upper bound on the sum over the right hand side. I claim:

$$\sum_y |\phi_T - \phi_T^y|^2 \leq \sum_y f \left(4 T \sum_i |P_y \phi_i|^2 \right) \leq N f \left(\frac{1}{N} \sum_y 4 T \sum_i |P_y \phi_i|^2 \right) \quad (22)$$

We know that we have an absolute (and not only a relative) maximum on the right hand side because in the whole area of interest $f' > 0$ and $f'' < 0$. More precisely, $f' > 0$ is true exactly as long as the number of steps T in Grover's algorithm is below the (fractional) optimum (which can be obtained from equation 6) of about $\pi/4\sqrt{N}$. The above inequality is saturated by Grover's algorithm as there the above optimal situation with equal angles between successive vectors is realized with this constant angle equal to φ given in equation 4. Thus we have established that Grover's algorithm is optimal.

4 Limits on parallelizing Grover's algorithm

Assume we have S (S for "space") identical oracles with exactly one marked element. Thus we can imagine that we have S identically constructed physical black boxes. In particular I assume that all oracles take the same time to answer a query.

We want to find the fastest way to obtain the marked element with a quantum computer that is allowed to use all these oracles (and may input entangled states to them). To formalize this in a "query complexity" way and assuming that querying will take much more time than the other operations in the algorithm, I only consider the "querying time", which is the time during the algorithm when any one of the oracles is working.

The quantum computer could query the individual oracles at any time, in particular it can start querying an oracle while another one is still running. In

the following I give an argument that without loss of potential power of the algorithm, we can assume that the oracles are always queried synchronously.

First imagine that we have just 2 oracles. We begin by querying the first one and while it is still working we start querying the second one. We can assume that while an oracle is working, only the oracle interacts with its input register. (If necessary, this can be assured by XORing the input state to a register reserved for the oracle.) Then it follows that the second oracle could be queried as soon as the first one, possibly by doing some preparatory gates ahead of time.

Now imagine we have S oracles. First consider the very first querying of an oracle in the algorithm. All the oracles which we start querying while the first one is still working can, by the above argument, be queried simultaneously with the first one. As there is no point in not using the other oracles during this time, we can assume that we start the algorithm by querying all S oracles simultaneously.

By applying the same argument to what happens after this first step, we get that we can assume that the second step also consists of querying all S oracles simultaneously. By iterating this we see that we can assume that the algorithm always queries all oracles simultaneously. Say it does this T times.

As before (equation 22) we have

$$\sum_y |\phi_T - \phi_T^y|^2 \leq N f \left(\frac{1}{N} 4T \sum_y \sum_i |P_y \phi_i|^2 \right) \leq 4T \sum_y \sum_i |P_y \phi_i|^2 \quad (23)$$

Where on the right I have also included the old unimproved result. Here ϕ_i is the QC state just before the S oracles are called (for the $i+1$ -st time). Now P_y is the projector onto those computational basis states where any oracle (possibly several ones) is queried on input y .

It is easy to see that:

$$|P_y \phi_i|^2 \leq \sum_{k=1}^S |P_y^k \phi_i|^2 \quad (24)$$

Here P_y^k is the projector onto those computational basis states where oracle number k is queried on input y . The inequality becomes an equality when there are no basis states in ϕ_i where several oracles are queried on input y .

From that we get (compare to equation 14):

$$\sum_y \sum_{i=0}^{T-1} |P_y \phi_i|^2 \leq \sum_{k=1}^S \sum_{i=0}^{T-1} \underbrace{\sum_y |P_y^k \phi_i|^2}_{=1} = S \cdot T \quad (25)$$

So we get the final result

$$\sum_y |\phi_T - \phi_T^y|^2 \leq N f\left(\frac{1}{N} 4T^2 S\right) \leq S \cdot 4T^2 \quad (26)$$

This shows that to get a certain success probability we can gain only a factor of $\sqrt{S}$ in T by using S oracles, but this is essentially the same performance as S independent Grover searches, each working on one S^{th} of the total search space. If N (the size of the search space) is divisible by S this is an exact statement, otherwise we still get an asymptotic statement.

5 Appendices

5.1 proof of inequality 8

The situation is as follows: a quantum system is in one of N pure states given by the normalized vectors ψ_y , $y = 0 \dots N-1$. The task is to find in which of these states it is by using any measurement procedure allowed by quantum theory. It is well known that if the ψ_y 's are not all pairwise orthogonal, this can only be done probabilistically (see e.g. [5]). Here we are interested in maximizing the average success probability when averaging over all N cases. We want to prove the following upper bound on this probability p :

$$2N - 2\sqrt{N}\sqrt{p} - 2\sqrt{N}\sqrt{N-1}\sqrt{1-p} \leq \sum_{y=0}^{N-1} |\psi_y - \psi|^2 \quad (27)$$

Measurement schemes can in general be such that when the measurement gives y then one is sure that the state was ψ_y , but then in general the answer can also be “I do not know”. Here we are not interested in such schemes. Because once we get an answer we can easily check whether it is correct, all we are interested in, is to maximize the probability of getting the right answer, irrespective of whether an unsuccessful measurement yields a wrong answer or “do not know”.

5.1.1 Grover's algorithm saturates the inequality

Here the ψ_y 's are the different final states (depending on the oracle) of the QC just before measurement and ψ is the state we get for the “zero”-oracle. After any number of iterations of Grover's algorithm these states can be written in terms of some p as:

$$\psi_y = \sqrt{p} |y\rangle + \sqrt{1-p} \frac{1}{\sqrt{N-1}} \sum_{y' \neq y} |y'\rangle \quad \text{and} \quad \psi = \frac{1}{\sqrt{N}} \sum_y |y\rangle \quad (28)$$

It is now easy to verify that inequality 27 is saturated.

5.1.2 proof of the inequality

In general the Hilbert space of the QC will have dimension $M > N$. We must assume this because it may in general gives the possibility of a measurement with a larger success probability. On the other hand a von Neumann measurement on such an enlarged space *is* really the best we can do ([5]). A von Neumann measurement is just a standard quantum measurement given by a hermitian operator or, essentially equivalently, by a set of (mutually orthogonal) eigenspaces which together span the whole Hilbert space. I write the ψ_y 's and ψ as follows in terms of some basis $|m\rangle$:

$$\psi_y = \sum_{m=0}^{M-1} c_m^y |m\rangle \quad \text{and} \quad \psi = \sum_{m=0}^{M-1} c_m |m\rangle \quad (29)$$

Without loss of generality we can assume that we measure in the basis $|m\rangle$. I denote by M_y the set of m 's which, when we obtain them from a measurement, will be interpreted as answer “ y ”. By p_y I denote the probability of thereby correctly identifying the state ψ_y :

$$p_y = \sum_{m \in M_y} |c_m^y|^2 \quad (30)$$

To prove inequality 27 we look for the minimal value its right hand side can assume for a given $p = 1/N \sum_y p_y$. We do the minimization in two steps, First we find the ψ_y (= the c_m^y 's) for which $|\psi_y - \psi|^2$ is minimal for a given fixed p_y and ψ . Using the Lagrange multiplier technique to find (tentative) extrema under some constraint we get the expression:

$$|\psi_y - \psi|^2 - \lambda_1 |\psi_y|^2 - \lambda_2 p_y = \sum_{m=0}^{M-1} |c_m^y - c_m|^2 - \lambda_1 \sum_{m=0}^{M-1} |c_m^y|^2 - \lambda_2 \sum_{m \in M_y} |c_m^y|^2 \quad (31)$$

where the first constraint (Lagrange multiplier λ_1) comes from the requirement that ψ_y be normalized and the second (Lagrange multiplier λ_2) because we want to minimize for fixed p_y . To find candidate extrema we set the derivatives of this expression with respect to the real and imaginary parts of c_m^y equal to zero. The well known “trick” that in this case one can formally treat the complex variable and its complex conjugate as the 2 independent real variables simplifies the calculation to obtain:

$$c_m = (1 - \lambda_1 - \lambda_2) c_m^y \quad \forall m \in M_y \quad \text{and} \quad c_m = (1 - \lambda_1) c_m^y \quad \forall m \notin M_y \quad (32)$$

By satisfying the constraints we get:

$$1 - \lambda_1 - \lambda_2 = \pm \frac{\sqrt{a_y}}{\sqrt{p_y}} \quad 1 - \lambda_1 = \pm \frac{\sqrt{1 - a_y}}{\sqrt{1 - p_y}} \quad \text{where} \quad a_y = \sum_{m \in M_y} |c_m|^2 \quad (33)$$

From that we get the following 4 candidates for a minimum:

$$|\psi_y - \psi|^2 = 2 - 2(\pm \sqrt{p_y} \sqrt{a_y} \pm \sqrt{1 - p_y} \sqrt{1 - a_y}) \quad (34)$$

Of course the minimum is reached when both signs are positive. Note that we do not have to worry that $|\psi_y - \psi|^2$ might be even smaller on some boundary of the parameter range we minimized over. This is because our parameter range (under the constraints) does not have a boundary, thus it is a bona fide manifold. Also our coordinate system is regular all over the manifold. By summing over y we get:

$$\sum_y |\psi_y - \psi|^2 \geq 2N - 2 \sum_y (\sqrt{p_y} \sqrt{a_y} + \sqrt{1 - p_y} \sqrt{1 - a_y}) \quad (35)$$

Now we look for the ψ (= the a_y 's) and the p_y 's for which this becomes minimal. We have the constraints $|\psi|^2 = \sum |c_m|^2 = \sum a_y = 1$ and $1/N \sum_y p_y = p$ for a fixed p . Again using the Lagrange multiplier technique we get $a_y = 1/N \forall y$ and $p_y = p \forall y$. Then:

$$\sum_{y=0}^{N-1} |\psi_y - \psi|^2 = 2N - 2\sqrt{N}\sqrt{p} - 2\sqrt{N}\sqrt{N-1}\sqrt{1-p} \quad (36)$$

This time the parameter range over which we have minimized does have a boundary. The boundary is reached when one of the a_y 's or p_y 's is either 0 or 1. Still we can show that we have really found a global minimum by showing that the second derivative is positive definite over the whole parameter range. To avoid having to adapt this argument to the situation with constraints, we can e.g. set $p_0 = Np - \sum_{y \neq 0} p_y$ and $a_0 = 1 - \sum_{y \neq 0} a_y$ and then check that the second derivative of the right hand side of equation 35 is always positive definite.

6 final remarks

I have here only considered oracles with the promise that there is exactly one marked element. It seems very plausible that the proof can be extended to oracles with any known(!) number of marked elements. The same may be true for the case where we have a non-uniform a priori probability for the different 1-yes oracles and we want to maximize the average success probability. Then one also has to consider a modified Grover algorithm.

Also it seems that by reading the proof carefully, one can establish that Grover's algorithm is essentially the only optimal algorithm.

As for any no-go theorem which claims implications for the physical world, we must be careful about the assumptions we made. Arguably the main assumption made here is that the time evolution of quantum states is exactly linear as of course it is in standard quantum theory. Most physicists think this is very likely.

I would like to thank Manny Knill and Daniel Gottesman for helpful discussions.

References

- [1] L. Grover
A fast quantum mechanical algorithm for database search
Proceedings, 28th Annual ACM Symposium on the Theory of Computing (STOC), May 1996, pages 212-219
also quant-ph/9605043
- [2] L. Grover
Quantum Mechanics helps in searching for a needle in a haystack
Phys. Rev. Letters, Volume 79, issue 2, pp. 325-328, July 14, 1997
also quant-ph/9706033
- [3] C. Bennett et al.
Strength and Weaknesses of Quantum Computing
SIAM Journal on Computing (special issue on quantum computing) volume 26, number 5, pages 1510-1523
also quant-ph/9701001
- [4] M. Boyer et al.
Tight bounds on quantum searching
Proceedings 4th Workshop on Physics and Computation, 36-43, 1996
also Fortschritte der Physik, Volume 46, Number 4-5, 493-505, 1998
quant-ph/9605034
- [5] Ch. Fuchs
Distinguishability and Accessible Information in Quantum Theory
Ph. D. Dissertation, University of New Mexico
quant-ph/9601020